

TITLE:

**A CHOLESTEROL-DRIVEN COMPOSITIONAL CROSSOVER TUNES THE
PHYSICAL STATE OF PHASE-SEPARATED LIPID MEMBRANES**

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ABSTRACT

Because lipid rafts regulate the spatial organization of receptors, signalling complexes, and membrane trafficking, understanding their cholesterol-dependent physical basis is increasingly important, particularly as altered plasma-membrane cholesterol organization is implicated in cancer progression^{1,2}. Here, we investigate supported lipid bilayers composed of a biomimetic quaternary mixture (DOPC/DPPC/SM/cholesterol) containing 7–33 mol% cholesterol. By combining AFM imaging, force spectroscopy, FRAP, calorimetry, and neutron spin-echo spectroscopy, we resolve membrane structure, mechanics, and dynamics across multiple length and time scales. We identify a compositional crossover near 25 mol% cholesterol that separates two distinct physical regimes. Below this threshold, the membrane exhibits strong phase separation, characterized by a large thickness mismatch, high line tension, pronounced mechanical contrast, and slow diffusion within liquid-ordered domains. Above the crossover, the thickness mismatch decreases rapidly, while the structural, mechanical, and dynamical contrast between liquid-ordered (Lo) and liquid-disordered (Ld) phases progressively collapses. AFM force spectroscopy reveals a convergence of rupture forces between the Lo and Ld phases, whereas FRAP measurements show increased molecular mobility within ordered domains, consistent with reduced interfacial constraints. Analysis of domain boundaries further reveals a minimum in line tension near the crossover, indicating reduced interfacial energy. Neutron spin-echo spectroscopy additionally identifies cholesterol-dependent changes in collective membrane fluctuations at nanometre length scales, demonstrating that the crossover extends from lateral membrane organization to collective dynamics. Together, these results show that cholesterol does not simply stabilize ordered domains, but drives a cholesterol-driven compositional crossover that reshapes the membrane free-energy landscape. This crossover shifts the system from a strongly heterogeneous, high-contrast state to a mechanically convergent regime with reduced interfacial energy and diminished phase

contrast. Our findings provide a unified multiscale framework linking membrane organization, mechanics, and collective dynamics, and suggest that biological membranes may operate near a compositional regime in which relatively small variations in cholesterol produce disproportionately large changes in raft organization and function. Defining this regime may therefore be relevant for understanding pathological alterations in membrane signalling and for developing strategies aimed at modulating raft-dependent cellular processes.

REFERENCES

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